Halbach magnetic assembly

X20 (2020) — English translation

The Halbach magnetic assembly is a special configuration of permanent magnets, characterized by the fact that the magnetic field on one of its sides is almost completely absent due to the special arrangement of the assembly elements. In this problem we explore this phenomenon.

Magnetic dipoles

A dipole is a point magnetic element (for example, a small loop with current. Its dipole moment is $I\vec{S}$, where I is the current running through the loop, and $\vec{S}$ is the oriented area). The field created by a magnetic dipole is described by the following formula:

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\left(\frac{3\vec{r}(\vec{m}\cdot\vec{r})}{r^5} - \frac{\vec{m}}{r^3}\right),$$

где m — дипольный момент, $\vec{r}$ — радиус-вектор точки в пространстве относительно диполя. $\mu_0 = 4\pi \cdot 10^{-7}$ H/m. The figure shows a dipole, a vector and the angle between them.

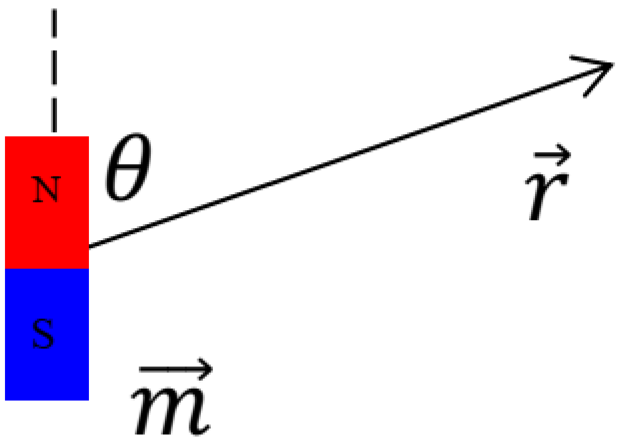

Fig. 1.

A1	The magnetic field weakens with increasing distance; for a given angle θ, find the dependence of the magnetic field B at a distance r from the dipole.	**0.50**

Magnetic washer

Let us consider a light magnet, which is a flat cylinder of radius R and thickness $h \ll R$ with a surface magnetic moment density $\vec{\sigma}$. The surface density vector is oriented along the disk axis.

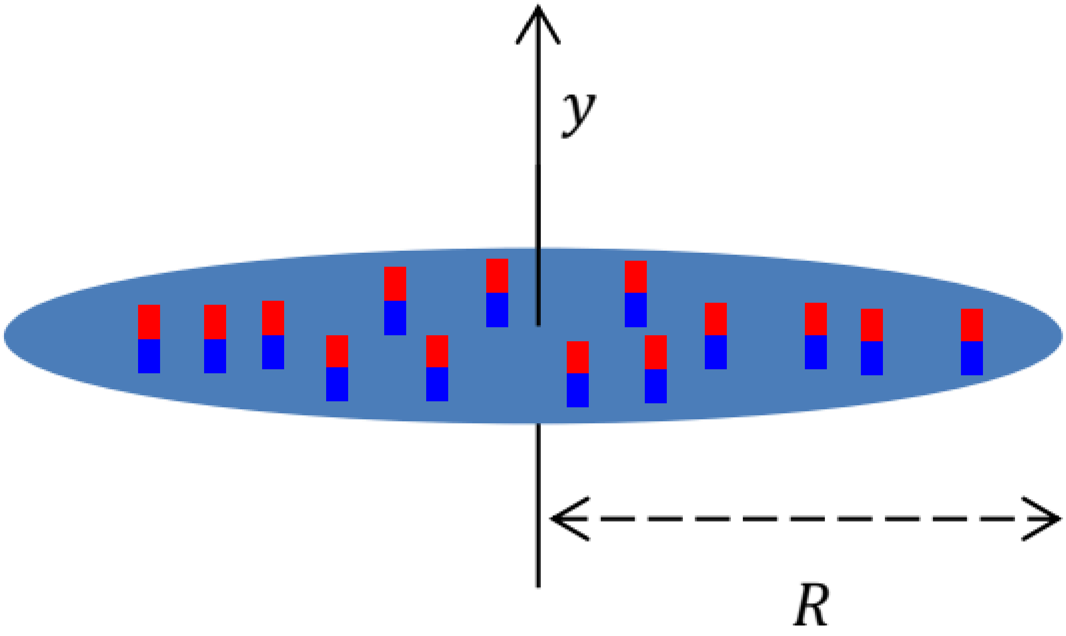

Fig. 2.

B1	Express the magnetic field $B(y)$ along an axis perpendicular to the magnet at a distance y from the center.	**1.50**

If you bring a small round magnet close to a metal refrigerator door, the magnet will be attracted to it with a force that depends on the size and type of material of the magnet, as well as the thickness of the door. At this point, consider the thickness of the door to be much larger than the linear dimensions of the magnet. Consider a cylindrical magnet with a volumetric dipole moment density $\rho = 1.05 \cdot 10^6 \frac{\text{Тл}\cdot\text{м}}{\text{Гн}}$ of thickness $t = 2$ мм and diameter $D = 20$ мм.

B2	Estimate the magnitude of the magnetic field near the surface of the magnet. Express your answer in terms of quantities $t,\ D,\ \rho,\ \mu_0$.	**0.50**

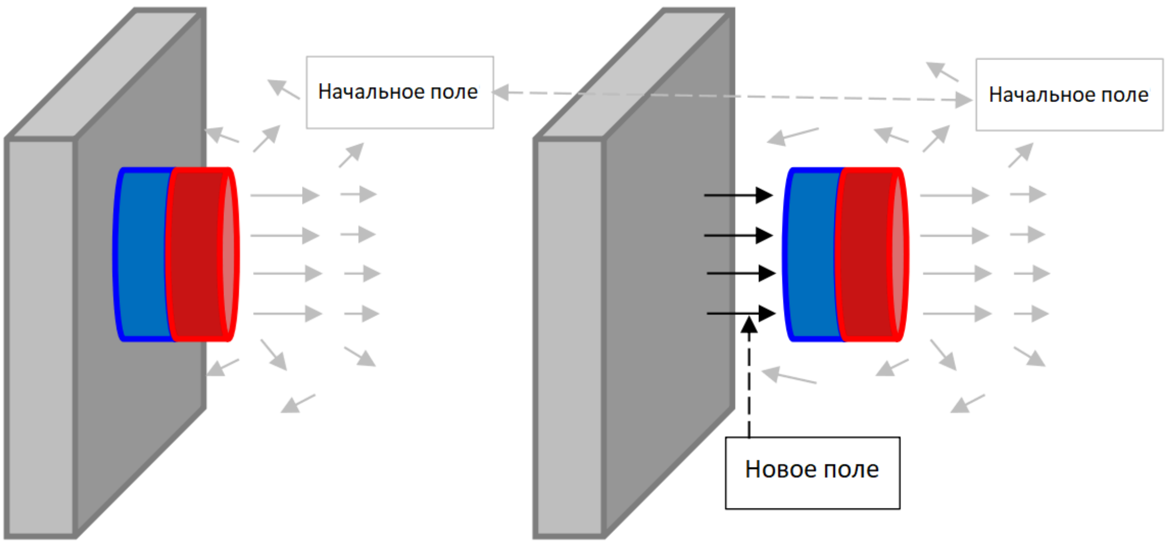

Fig. 3.

To determine the force of interaction between the magnet and the refrigerator door, it is necessary to use the law of conservation of energy. When a magnet is pulled away from a door, a field is created between them that is approximately equal to the field on the other side of the magnet. The rest of the field (including what's inside the door) remains almost unchanged. The expression for the volumetric magnetic field energy density in air is: $w = \frac{B^2}{2\mu_0}$.

B3	Obtain the expression and numerical value of the interaction force F_0 between the door and the magnet pressed to it, and also calculate the pressure P_0 of the magnet on the door.	**0.50**

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In order to obtain the expression of the magnetic field in the Halbach assembly, we find the field of a long row of magnets, as shown in the figure. Linear density of the dipole moment of the series ρ_L, direction - along the axis y.

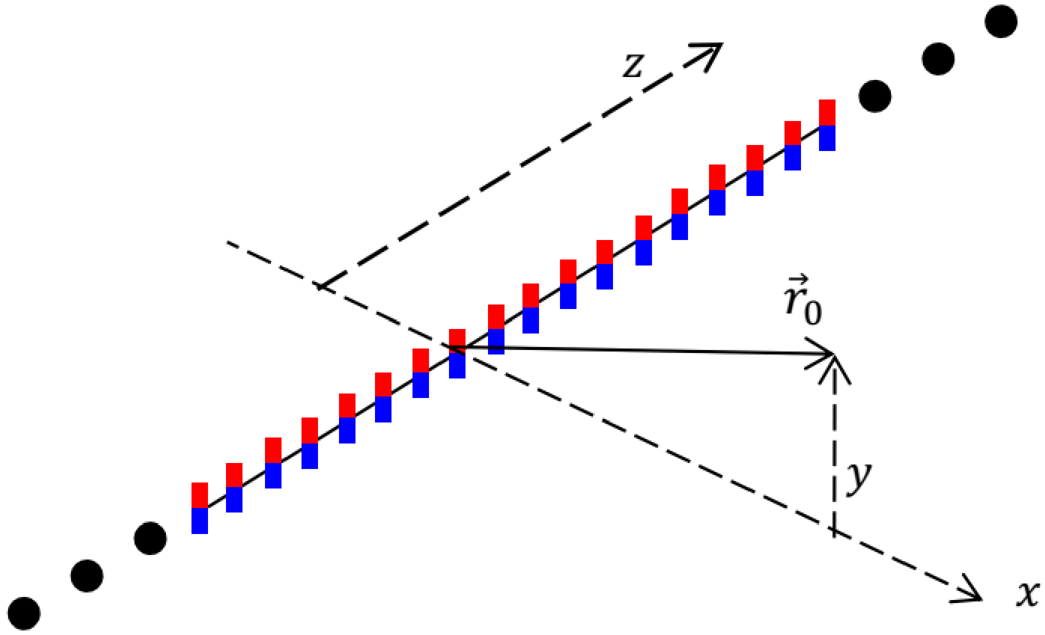

Fig. 4.

C1	Write down an expression for the field $\vec{B}(\vec{r}_0, y)$ that is created by a series of magnets. (For convenience, the field is expressed as both $\vec{r}_0$ and y, although technically it is $y = (\vec{r}_0)_y$.)	**2.00**

In a planar Halbach magnetic assembly, the polarization direction of a small area element rotates continuously. Its position changes in accordance with the formula:

$$\beta(x, z) = \beta_0 + k \cdot x,\ k = 2\pi/\lambda,\ t \ll \lambda$$

Здесь β — это угол между направлением диполя и перпендикуляром к плоскости, этот угол вращается в плоскости xy. Длина λ — это шаг сборки, а t - thickness of the assembly.

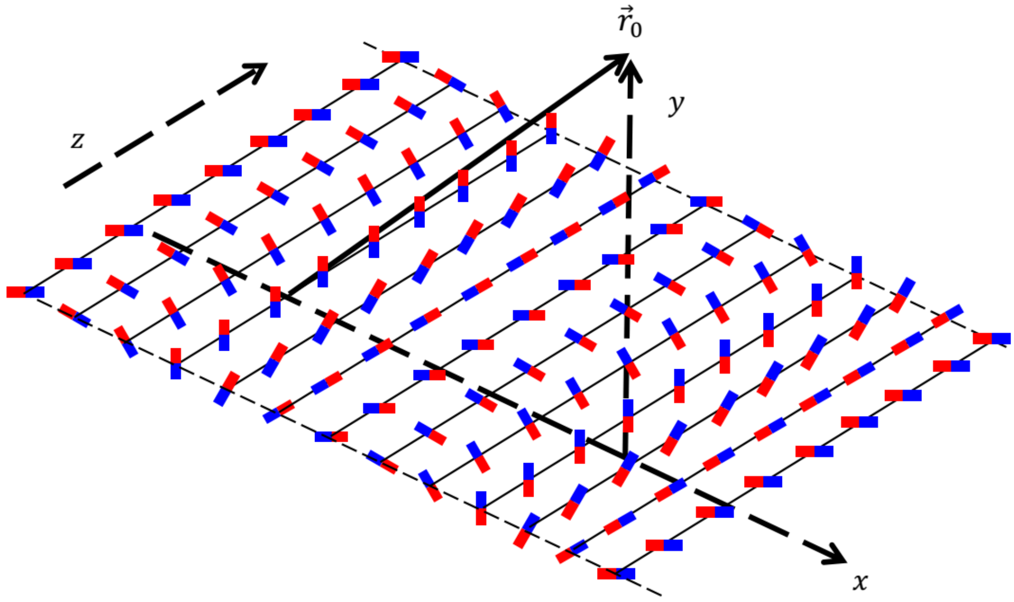

Fig. 5.

C2	Find the magnetic field on both sides of the assembly. Give the answer in the form of some integral.	**1.00**

One of these integrals is needed to solve this point:

$$\int_{-\pi/2}^{\pi/2} dx \cdot \cos(2x) \cdot \cos(c \cdot \tan x) = \frac{c \cdot \pi}{e^c} = \int_{-\pi/2}^{\pi/2} dx \cdot \sin(2x) \cdot \sin(c \cdot \tan x)$$

$$\frac{\pi}{e^d} = \int_{-\infty}^{\infty} dx \cdot \frac{\cos x}{d^2 + x^2}$$

$$\frac{\pi}{a \cdot e^a} = \int_{-\infty}^{\infty} dx \cdot \frac{2x \cdot \sin(2x)}{(a^2 + x^2)^2}$$

$$\frac{(b+1)\pi}{e^b} = \int_{-\infty}^{\infty} dx \cdot \frac{2b^3 x \cdot \cos(x)}{(b^2 + x^2)^2}$$

C3	Show that on one side of an ideal assembly the magnetic field tends to zero.	**1.00**

C4	Write down the expression for the field on the other side.	**1.00**

C5	Based on the field expression, find the average pressure P of such a magnet on the refrigerator door. Take the following parameters: thickness $t = 0.5$ мм, volumetric density of the magnetic dipole $\rho = 2 \cdot 10^5 \frac{\text{Тл}\cdot\text{м}}{\text{Гн}}$, assembly step $\lambda = 5$ мм.	**1.50**

C6	Find the ratio between the pressure created by a Halbach magnetic assembly and the pressure created by a conventional magnet of the same material, with the same radius and thickness. Here, too, we should neglect the effects on the perimeter of the circle and the thickness of the magnet.	**0.50**

You might think that nothing more innovative than refrigerator magnets would use Halbach assembly. However, the Hendo company has developed an entire hoverboard based on it (the same one that was in the movie "Back to the Future"). Using Halbach's ring assembly of electromagnets, Hendo achieved enough lift to lift an average-sized person off the ground! But only over copper.

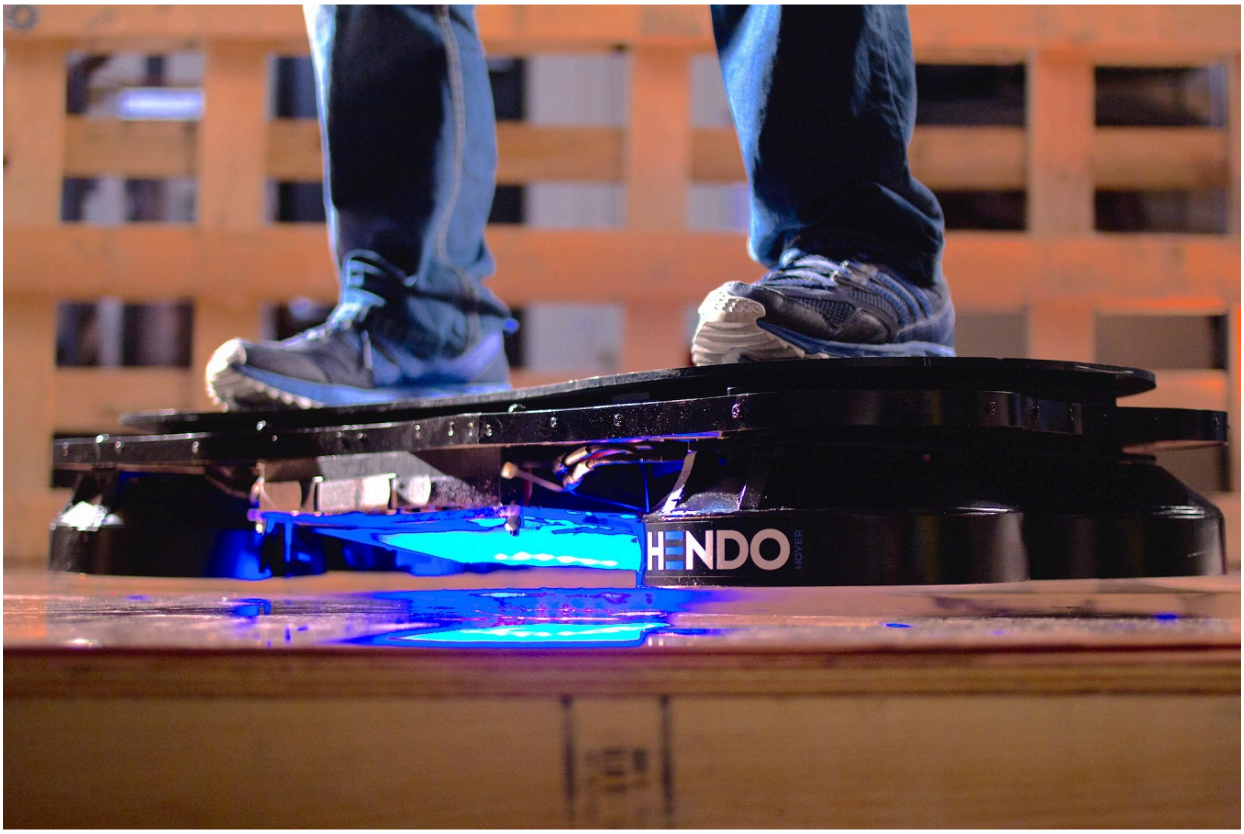

Fig. 6.

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